

Ill-Conditioning of the Quadratic Penalty Method and Its Removal by the Augmented Lagrangian

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Seminar paper in Numerical Optimisations, Data Science and Artificial Intelligence

Professor: Izudin Džafić, academic year 2025/26

Abstract: The quadratic penalty method replaces an equality-constrained problem by a sequence of unconstrained problems in which the penalty parameter μ grows without bound. The Hessian of the penalty function then has m eigenvalues of order μ , so Newton's method is applied to increasingly ill-conditioned matrices. This paper measures that effect with an implementation written from scratch in C++20 on the natID framework: a modified Newton method with exact second derivatives solves every subproblem, and the condition number of the Hessian is recorded after every outer iteration. On five test problems, including an algebraic ellipse fit under a normalisation constraint, the condition number in the ∞ -norm grows with a log-log slope between 0.9995 and 1.0001 and reaches 1.3×10^{12} to 8.2×10^{12} at $\mu = 10^{12}$. From $\mu = 10^6$ to 10^8 onward the inner Newton method can no longer reduce the gradient below its tolerance, because the attainable gradient accuracy in floating point grows in proportion to μ . The augmented Lagrangian method with a fixed $\mu = 10$ meets the same stopping tests with condition numbers between 14 and 84, needs 92 instead of 127 Newton iterations over all problems, and reduces the constraint violation of the curve fit to 3.3×10^{-16} . A parameter study shows that this advantage disappears when the augmented Lagrangian is run with a large μ , where it inherits the conditioning of the penalty method.

Index Terms: quadratic penalty method, augmented Lagrangian, method of multipliers, Newton's method, condition number, equality constraints, conic fitting

I. INTRODUCTION

Penalty methods are among the oldest ways of handling constraints in optimisation [1], [2]. They replace a constrained problem by unconstrained problems that any unconstrained solver can handle, and they are simple to implement. Their weakness is equally well known: feasibility is reached only in the limit of an infinite penalty parameter, and the Hessian of the penalty function becomes ill-conditioned along the way [3], [4], [5]. The augmented Lagrangian method, or method of multipliers, introduced independently by Hestenes [6] and Powell [7], adds an explicit multiplier estimate and reaches feasibility with a bounded penalty parameter [8]. It is the basis of production codes such as LANCELOT [9] and ALGENCAN [10].

Textbooks state these properties, but general-purpose solvers do not show them: the condition number of the matrix factorised in each Newton step is internal to the solver. The goal

of this work is to make the conditioning of every step visible and to relate it to the behaviour of Newton's method. All derivatives, the inner Newton solver, the outer loops and the condition number estimates are written from scratch, so every quantity can be recorded.

The contributions are:

- a common implementation of the quadratic penalty and augmented Lagrangian methods on top of one modified Newton solver, so that differences in the results come from the outer method only;
- measurements of κ_∞ , κ_1 and κ_2 of the Hessian for μ from 1 to 10^{12} on five problems, together with the accuracy and the inner-solver status at each step;
- a comparison of feasibility, accuracy and work at default settings, a study of the penalty growth factor and of the fixed augmented Lagrangian parameter, and a robustness study on the curve-fitting problem;
- an interactive cross-platform application that reproduces all experiments.

Section II states the problem and the two methods, Section III the inner solver and the condition number estimates, Section IV the test problems and Section V the implementation. Results are reported in Section VI and discussed in Section VII.

II. PROBLEM AND METHODS

A. Optimality conditions

We consider

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad h(x) = 0, \quad (1)$$

with $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$ twice continuously differentiable and $m < n$. With the Lagrangian $L(x, \lambda) = f(x) + \lambda^\top h(x)$ and the constraint Jacobian $J(x) = \nabla h(x)^\top \in \mathbb{R}^{m \times n}$, a regular local minimiser x^* satisfies the Karush-Kuhn-Tucker (KKT) conditions

$$\nabla f(x^*) + J(x^*)^\top \lambda^* = 0, \quad h(x^*) = 0. \quad (2)$$

We assume that $J(x^*)$ has full row rank and that $Z^\top \nabla_{xx}^2 L(x^*, \lambda^*) Z \succ 0$, where the columns of Z span the null space of $J(x^*)$.

Algorithm 1 Quadratic penalty method

Input: $x_0, \mu_0 > 0$, growth factor $\rho > 1$, $\mu_{\max}$

```

1:  $x \leftarrow x_0, \mu \leftarrow \mu_0$ 
2: for  $k = 0, 1, \dots, k_{\max} - 1$  do
3:    $x_k \leftarrow \text{NEWTON}(Q(\cdot; \mu), x)$   $\triangleright$  warm start
4:    $\lambda_k \leftarrow \mu h(x_k)$ 
5:   if  $\|h(x_k)\|_\infty \leq \varepsilon_{\text{feas}}$  and  $\|\nabla f(x_k) + J_k^\top \lambda_k\|_\infty \leq \varepsilon_{\text{opt}}$ 
     then
6:     return  $x_k, \lambda_k$ 
7:   end if
8:   if  $\rho\mu > \mu_{\max}$  then return  $x_k, \lambda_k$ 
9:   end if
10:   $x \leftarrow x_k, \mu \leftarrow \rho\mu$ 
11: end for

```

B. Quadratic penalty method

The quadratic penalty function is

$$Q(x; \mu) = f(x) + \frac{\mu}{2} \sum_{i=1}^m h_i(x)^2 = f(x) + \frac{\mu}{2} \|h(x)\|_2^2, \quad (3)$$

with gradient and Hessian

$$\nabla Q(x; \mu) = \nabla f(x) + J(x)^\top (\mu h(x)), \quad (4)$$

$$\begin{aligned} \nabla^2 Q(x; \mu) &= \nabla^2 f(x) + \sum_{i=1}^m \mu h_i(x) \nabla^2 h_i(x) \\ &\quad + \mu J(x)^\top J(x). \end{aligned} \quad (5)$$

For an increasing sequence $\mu_k \rightarrow \infty$ the method computes $x_k \approx \arg \min_x Q(x; \mu_k)$, starting each minimisation from x_{k-1} (Algorithm 1). Comparing $\nabla Q(x_k; \mu_k) = 0$ with (2) gives the multiplier estimate

$$\lambda_k = \mu_k h(x_k), \quad \text{so} \quad h(x_k) = \frac{\lambda_k}{\mu_k} \approx \frac{\lambda^*}{\mu_k}. \quad (6)$$

The constraint violation, and with it the distance to x^* , therefore decreases only like $1/\mu_k$ unless $\lambda^* = 0$.

Substituting $\mu_k h_i(x_k) = \lambda_{k,i}$ into (5) shows the structure that causes ill-conditioning:

$$\nabla^2 Q(x_k; \mu_k) = \nabla_{xx}^2 L(x_k, \lambda_k) + \mu_k J_k^\top J_k. \quad (7)$$

The rank- m term $\mu_k J_k^\top J_k$ contributes m eigenvalues of order $\mu_k \sigma_i(J_k)^2$, while the remaining $n - m$ eigenvalues approach those of the reduced Hessian $Z^\top \nabla_{xx}^2 L Z$ and stay bounded [3]. Hence

$$\kappa_2(\nabla^2 Q(x_k; \mu_k)) \approx \frac{\mu_k \sigma_{\max}(J_k)^2}{\lambda_{\min}(Z^\top \nabla_{xx}^2 L Z)} = O(\mu_k). \quad (8)$$

C. Augmented Lagrangian method

The augmented Lagrangian adds a linear multiplier term to (3):

$$\mathcal{L}_A(x; \lambda, \mu) = f(x) + \lambda^\top h(x) + \frac{\mu}{2} \|h(x)\|_2^2, \quad (9)$$

Algorithm 2 Augmented Lagrangian method

Input: $x_0, \mu > 0$

```

1:  $x \leftarrow x_0, \lambda \leftarrow 0$ 
2: for  $k = 0, 1, \dots, k_{\max} - 1$  do
3:    $x_k \leftarrow \text{NEWTON}(\mathcal{L}_A(\cdot; \lambda, \mu), x)$ 
4:    $\lambda \leftarrow \lambda + \mu h(x_k)$ 
5:   if  $\|h(x_k)\|_\infty \leq \varepsilon_{\text{feas}}$  and  $\|\nabla f(x_k) + J_k^\top \lambda\|_\infty \leq \varepsilon_{\text{opt}}$ 
     then
6:     return  $x_k, \lambda$ 
7:   end if
8:    $x \leftarrow x_k$   $\triangleright$  optionally update  $\mu$  by (15)
9: end for

```

$$\nabla \mathcal{L}_A = \nabla f(x) + J(x)^\top (\lambda + \mu h(x)), \quad (10)$$

$$\begin{aligned} \nabla^2 \mathcal{L}_A &= \nabla^2 f(x) + \sum_{i=1}^m (\lambda_i + \mu h_i(x)) \nabla^2 h_i(x) \\ &\quad + \mu J(x)^\top J(x). \end{aligned} \quad (11)$$

The quadratic penalty function is the special case $Q(x; \mu) = \mathcal{L}_A(x; 0, \mu)$, and the implementation uses one class for both. After $x_k \approx \arg \min_x \mathcal{L}_A(x; \lambda_k, \mu_k)$ the multipliers are updated by

$$\lambda_{k+1} = \lambda_k + \mu_k h(x_k), \quad (12)$$

which makes (10) equal to $\nabla f(x_k) + J_k^\top \lambda_{k+1}$, the stationarity part of (2). The violation $h(x_k) = (\lambda_{k+1} - \lambda_k)/\mu_k$ now tends to zero because the multipliers converge, not because μ grows. For μ above a threshold $\bar{\mu}$ and λ_k close to λ^* there is a constant M with [5], [8]

$$\|x_k - x^*\| \leq \frac{M}{\mu} \|\lambda_k - \lambda^*\|, \quad (13)$$

$$\|\lambda_{k+1} - \lambda^*\| \leq \frac{M}{\mu} \|\lambda_k - \lambda^*\|, \quad (14)$$

so the multipliers converge linearly with rate M/μ for a fixed μ , and the Hessian (11) keeps a condition number of order μ that does not grow with k .

By default μ is fixed (Algorithm 2). An optional adaptive rule increases it only when feasibility stalls:

$$\mu_{k+1} = \begin{cases} \min(\rho_A \mu_k, \mu_{\max}^A), & \|h(x_k)\|_\infty > \eta \|h(x_{k-1})\|_\infty, \\ \mu_k, & \text{otherwise,} \end{cases} \quad (15)$$

with $\rho_A = 5$, $\eta = 1/4$ and $\mu_{\max}^A = 10^8$.

Both methods stop when $\|h(x_k)\|_\infty \leq \varepsilon_{\text{feas}} = 10^{-8}$ and the KKT residual with the current multiplier estimate satisfies $\|\nabla f(x_k) + J_k^\top \lambda\|_\infty \leq \varepsilon_{\text{opt}} = 10^{-6}$, after at most $k_{\max} = 30$ outer iterations. The penalty method starts from $\mu_0 = 1$ with $\rho = 10$ and $\mu_{\max} = 10^{12}$; the augmented Lagrangian uses $\mu = 10$.

III. INNER SOLVER AND CONDITION NUMBERS**A. Modified Newton method**

Both outer methods minimise a twice differentiable merit function Φ , either Q or $\mathcal{L}_A$, with the same Newton method. At an iterate x with $g = \nabla \Phi(x)$ and $H = \nabla^2 \Phi(x)$:

Hessian modification. If a Cholesky factorisation of H fails, the method uses $H + \tau I$ with

$$\tau_0 = \max(10^{-3}, 10^{-3}\|H\|_\infty), \quad \tau \leftarrow 10\tau, \quad (16)$$

until the factorisation succeeds [5, Alg. 3.3].

Newton direction. The step solves $(H + \tau I)p = -g$ with the LU based solver of natID. If $g^\top p \geq 0$ the steepest descent direction $p = -g$ is used.

Line search. The step length starts at $\alpha = 1$ and is halved, at most 60 times, until the Armijo condition

$$\Phi(x + \alpha p) \leq \Phi(x) + c_1 \alpha g^\top p, \quad c_1 = 10^{-4}, \quad (17)$$

holds.

Rounding regime. When the predicted decrease is below the resolution of Φ ,

$$-g^\top p \leq 64 \varepsilon_{\text{mach}} \max(1, |\Phi(x)|), \quad (18)$$

condition (17) cannot separate progress from rounding noise. The full step is then accepted only if it reduces $\|\nabla\Phi\|_\infty$; otherwise the method stops with the status *stalled*.

The iteration stops when $\|g\|_\infty \leq 10^{-9}$ or after 100 iterations. For each subproblem the solver reports the iteration count, the final gradient norm, the largest shift τ and the largest relative residual $\|(H + \tau I)p + g\|_\infty / \|g\|_\infty$ of the linear solves.

B. Condition number estimates

After each subproblem the condition number of the Hessian of the merit function that was minimised is evaluated at x_k . For $p \in \{1, \infty\}$

$$\kappa_p(A) = \|A\|_p \|A^{-1}\|_p, \quad (19)$$

$$\|A\|_\infty = \max_i \sum_j |a_{ij}|, \quad \|A\|_1 = \max_j \sum_i |a_{ij}|,$$

with A^{-1} from the natID matrix inverse. The spectral condition number of the symmetric Hessian is

$$\kappa_2(A) = \frac{\max_i |\lambda_i(A)|}{\min_i |\lambda_i(A)|}, \quad (20)$$

with eigenvalues from the cyclic Jacobi method [11, Sec. 8.5], since the SDK offers no eigenvalue or singular value routine. The norms are equivalent up to the dimension, $n^{-1}\kappa_2(A) \leq \kappa_\infty(A) \leq n\kappa_2(A)$ [12], so all three show the same growth rate. Unless stated otherwise, results use κ_∞ .

IV. TEST PROBLEMS

Table I lists the problems. Four have closed-form solutions; the reference for the curve fit is computed independently from an eigenvalue problem.

TABLE I
TEST PROBLEMS AND REFERENCE SOLUTIONS

Problem	n	m	x^*	λ^*
Conic fit	6	1	eigenvector of M	$-\sigma_{\min}(M)$
Circle projection	2	1	$(2, 1)/\sqrt{5}$	$\sqrt{5} - 1$
N&W Example 17.1 [5]	2	1	$(-1, -1)$	$1/2$
HS 7 [13]	2	1	$(0, \sqrt{3})$	$1/(2\sqrt{3})$
Equality QP	4	2	KKT system	KKT system

A. Conic fit under a normalisation constraint

A conic through a point (u, v) satisfies $d(u, v)^\top \theta = 0$ with $d(u, v) = (u^2, uv, v^2, u, v, 1)^\top$. Given data points, the algebraic least squares fit [14] is

$$\min_{\theta \in \mathbb{R}^6} \theta^\top M \theta \quad \text{s.t.} \quad \theta^\top \theta - 1 = 0, \quad M = \frac{1}{N} \sum_{i=1}^N d_i d_i^\top. \quad (21)$$

Without the constraint, $\theta = 0$ would be the trivial minimiser. The unit-norm constraint is the simplest normalisation; other choices, such as the ellipse-specific constraint of Fitzgibbon et al. [15], lead to generalised eigenvalue problems. The derivatives are $\nabla f = 2M\theta$, $\nabla^2 f = 2M$, $\nabla h = 2\theta$ and $\nabla^2 h = 2I$. The KKT conditions $2M\theta + 2\lambda\theta = 0$ give $M\theta^* = -\lambda^*\theta^*$, so θ^* is a unit eigenvector for the smallest eigenvalue $\sigma_{\min}(M)$ and $\lambda^* = -\sigma_{\min}(M)$, which is also the optimal residual f^* . Because θ and $-\theta$ describe the same conic, the error is $\min(\|\theta - \theta^*\|, \|\theta + \theta^*\|)$.

The $N = 40$ data points lie on an ellipse with centre $(1.5, -0.5)$, semi-axes 3 and 1.5 and rotation 30° , at parameters $t_i = 2\pi(i + \frac{1}{2}r_i)/N$ with r_i uniform on $(0, 1)$, and are perturbed by Gaussian noise with standard deviation 0.05. Noise is generated with the Mersenne Twister and the Box-Muller transform, seed 20154, so the data are identical on every platform. Before the design vectors are formed the points are centred and scaled to a mean distance of $\sqrt{2}$ from the origin [16], which keeps M well scaled: for the default data its eigenvalues range from 7.34×10^{-4} to 4.39. The starting point $\theta_0 = (1, 0, 1, 0, 0, -1)/\sqrt{3}$ is the unit circle in normalised coordinates.

B. Analytic problems

Circle projection: $\min (x_1 - 2)^2 + (x_2 - 1)^2$ subject to $x_1^2 + x_2^2 - 1 = 0$, from $x_0 = (-1, 1.5)$.

Nocedal and Wright, Example 17.1: $\min x_1 + x_2$ subject to $x_1^2 + x_2^2 - 2 = 0$, from $x_0 = (1, 0.5)$. For small μ the penalty function is not convex, so the Hessian modification (16) is needed.

Hock-Schittkowski problem 7: $\min \ln(1 + x_1^2) - x_2$ subject to $(1 + x_1^2)^2 + x_2^2 - 4 = 0$, from $x_0 = (2, 2)$.

Equality-constrained QP: $\min \frac{1}{2}x^\top A x - b^\top x$ subject to $Cx = d$ with $n = 4$, $m = 2$, a symmetric positive definite tridiagonal A and a full-rank C . The reference solves

$$\begin{bmatrix} A & C^\top \\ C & 0 \end{bmatrix} \begin{bmatrix} x^* \\ \lambda^* \end{bmatrix} = \begin{bmatrix} b \\ d \end{bmatrix}. \quad (22)$$

V. IMPLEMENTATION

The application is written in C++20 with the natID framework [17], which provides the user interface, the dense matrix type with LU solve and inverse, and the build and installer tooling for Windows, macOS and Linux. No platform-specific code is used.

A. Design

The numerical core has no user-interface dependencies and follows the SOLID principles. A problem implements the interface `IProblem`, which supplies f , ∇f , $\nabla^2 f$, h , J and $\nabla^2 h_i$; optional capabilities such as a plotting window for two-dimensional problems or the data of a curve fit are separate small interfaces. Both methods implement `IConstrainedSolver` and share `NewtonSolver`. Every outer iteration is stored in a `SolveHistory` record with μ , the multipliers, f , $\|h\|$, the KKT residual, the condition number, the Newton diagnostics and the errors with respect to the reference solution. All plots, tables and exports are produced from these records, and a new test problem needs one entry in a registry and no change to solver code. Cholesky factorisation and the Jacobi eigenvalue method are implemented in the project because the SDK does not provide them.

B. Verification

A console program checks the core with 158 tests that all pass: analytic gradients, Hessians and Jacobians against central finite differences, both methods against the reference solutions (penalty error below 10^{-5} , augmented Lagrangian error below 10^{-6} and multiplier error below 10^{-5}), a log-log slope of the penalty condition number between 0.8 and 1.2, and a ratio below 10 between the largest and smallest augmented Lagrangian condition numbers.

C. Application

Fig. 1 shows the application. A navigation rail on the left selects five pages: *Summary* with findings generated from the histories and a scorecard of both methods, *Landscape* with contours of Q or $\mathcal{L}_A$ and the Newton path, *Convergence*, *Conditioning* and *Data* with the iteration log. The inspector on the right selects the problem and holds sliders for all parameters. Computations run on a background thread and can be stopped; with automatic re-run enabled, every change of a slider repeats the comparison. Figures are exported as PDF or SVG and data as CSV into a time-stamped folder.

VI. RESULTS

All experiments were run with a console program that compiles the numerical core of the application without changes, on an Intel Core i5-1345U with 32 GB of memory under Windows 11 (MSVC, Release build). Unless noted, the default settings of Section II apply.



Fig. 1. Summary page of the application after a run on the conic fit: findings, scorecard with the better value marked, and condition number and constraint violation per outer iteration.

A. Comparison at default settings

Table II summarises one run of both methods on every problem. Both methods satisfy the stopping tests on all five problems. The penalty method needs 44 subproblems and 127 Newton iterations in total, the augmented Lagrangian 28 subproblems and 92 Newton iterations. The largest condition number of the penalty method lies between 8.2×10^5 and 5.7×10^9 and is reached at the final μ , which ranges from 10^5 to 10^9 . The augmented Lagrangian stays between 16 and 84.

Fig. 2 compares the methods on the curve fit. The penalty violation falls by exactly one decade per outer iteration, from 7.34×10^{-4} at $\mu = 1$ to 7.34×10^{-9} at $\mu = 10^5$. These values equal $|\lambda^*|/\mu = \sigma_{\min}(M)/\mu$, as predicted by (6). The first augmented Lagrangian subproblem, with $\lambda_0 = 0$ and $\mu = 10$, is identical to the second penalty subproblem and ends with the violation 7.34×10^{-5} . After one multiplier update the violation drops to 3.3×10^{-16} . Its condition number is 80 in both subproblems, while the penalty Hessian ends at 8.2×10^5 . The fitted conics of both methods coincide with each other and follow the noisy data (Fig. 3).

B. Growth of the condition number

To isolate the conditioning from the stopping test, the penalty method was run with $\varepsilon_{\text{feas}} = \varepsilon_{\text{opt}} = 0$ until $\mu = 10^{12}$, and the augmented Lagrangian for 12 outer iterations with $\mu = 10$. Fig. 4 shows the result. For every problem the penalty condition number follows a line of slope one: least squares fits over $\mu \in [10^2, 10^{12}]$ give slopes between 0.9995 and 1.0001. At $\mu = 10^{12}$, κ_{∞} lies between 1.3×10^{12} and 8.2×10^{12} and κ_2 between 8.9×10^{11} and 8.0×10^{12} . The ratio κ_{∞}/μ settles to a constant for each problem, 8.23 for the conic fit. For this problem (8) can be evaluated exactly: $J = 2\theta^T$ gives $\mu J^T J$ the eigenvalue 4μ , and $\nabla_{xx}^2 L = 2(M - \sigma_{\min} I)$ restricted to the orthogonal complement of θ^* has the smallest eigenvalue $2(\sigma_2(M) - \sigma_{\min}(M)) = 0.6864$. The prediction $\kappa_2 \approx 4\mu/0.6864 = 5.827\mu$ agrees with the measured $\kappa_2 = 5.827 \times 10^4$ at $\mu = 10^4$ to four digits. For symmetric

TABLE II
BOTH METHODS AT DEFAULT SETTINGS. QP: QUADRATIC PENALTY ($\mu_0 = 1, \rho = 10$); AL: AUGMENTED LAGRANGIAN ($\mu = 10$). SLOPE: LEAST SQUARES SLOPE OF $\log \kappa_\infty$ AGAINST $\log \mu$ FOR $\mu \geq 10^2$.

Problem	Method	Subproblems	Newton iter.	$\ h(x_k)\ _\infty$	$\ x_k - x^*\ _2$	$\max_k \kappa_\infty$	final μ	Slope
Conic fit	QP	6	14	7.3×10^{-9}	3.7×10^{-9}	8.2×10^5	1×10^5	1.000
	AL	2	8	3.3×10^{-16}	3.7×10^{-16}	8.0×10^1	1×10^1	n/a
Circle projection	QP	10	33	1.2×10^{-9}	6.2×10^{-10}	1.3×10^9	1×10^9	0.999
	AL	9	26	1.1×10^{-9}	5.4×10^{-10}	1.6×10^1	1×10^1	n/a
N&W Example 17.1	QP	9	33	5.0×10^{-9}	1.8×10^{-9}	8.0×10^8	1×10^8	1.000
	AL	5	26	1.1×10^{-9}	3.9×10^{-10}	8.4×10^1	1×10^1	n/a
HS 7	QP	9	33	2.9×10^{-9}	8.3×10^{-10}	3.8×10^8	1×10^8	1.000
	AL	4	24	3.1×10^{-9}	9.0×10^{-10}	3.9×10^1	1×10^1	n/a
Equality QP	QP	10	14	1.2×10^{-9}	8.5×10^{-10}	5.7×10^9	1×10^9	1.000
	AL	8	8	4.6×10^{-9}	2.8×10^{-9}	6.0×10^1	1×10^1	n/a

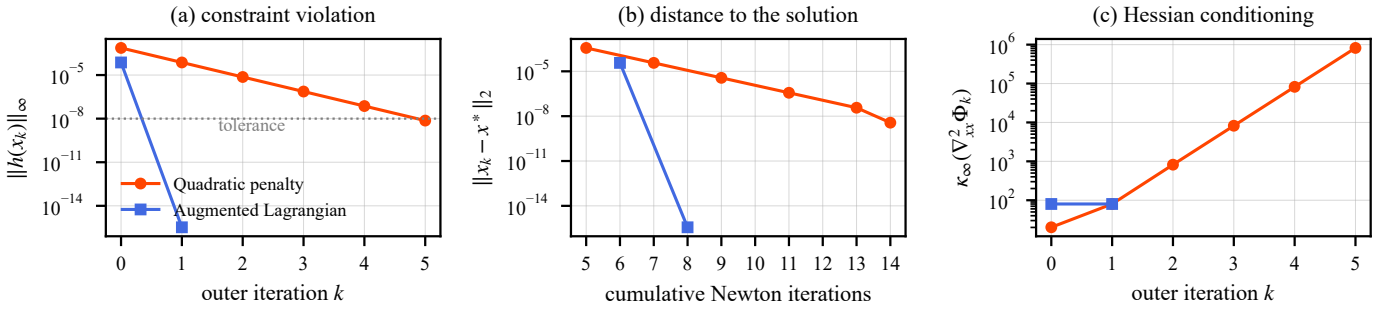


Fig. 2. Conic fit at default settings. (a) Constraint violation per outer iteration with the feasibility tolerance 10^{-8} . (b) Distance to the reference solution against the cumulative number of Newton iterations. (c) Condition number of the Hessian of the merit function Φ_k minimised in outer iteration k .

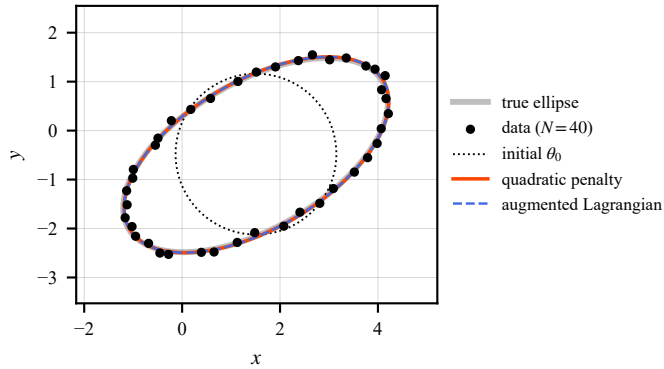


Fig. 3. Conic fit to 40 noisy points. The initial conic is the unit circle in normalised coordinates; the conics found by both methods are indistinguishable.

matrices $\kappa_1 = \kappa_\infty$, and the computed values agree to at least four significant digits.

The augmented Lagrangian values do not change with k . Over 12 outer iterations the largest and smallest κ_∞ differ by less than 17% on every problem (13.8 to 16.1 for the circle projection, 81.0 to 84.0 for Example 17.1), and they are close to the penalty values at the same $\mu = 10$. The augmented Lagrangian is therefore not better conditioned for a given μ ; it is better conditioned because it does not need a large μ .

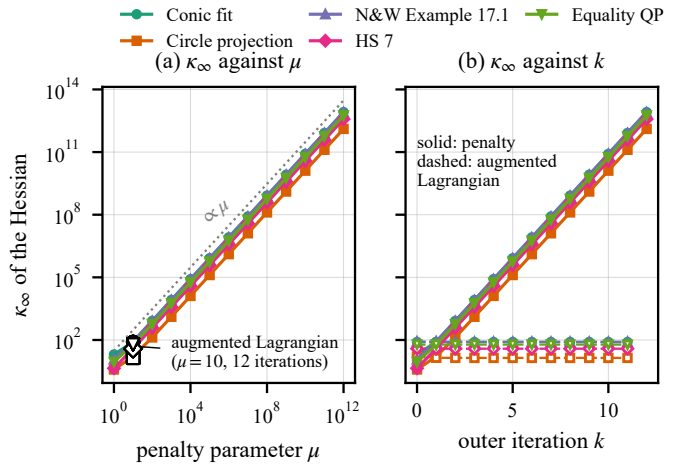


Fig. 4. Condition number κ_∞ of the merit Hessian. (a) Against the penalty parameter; the penalty method follows $\kappa_\infty \propto \mu$, and all 12 augmented Lagrangian iterations with $\mu = 10$ fall on one point per problem. (b) Against the outer iteration.

C. Effect on the Newton method

Fig. 5 shows what the growing condition number does to the inner solver. The distance to the solution keeps decreasing as c/μ over the whole range, with $\mu\|x_k - x^*\|$ converging to a constant for each problem (0.618 for the circle projection).

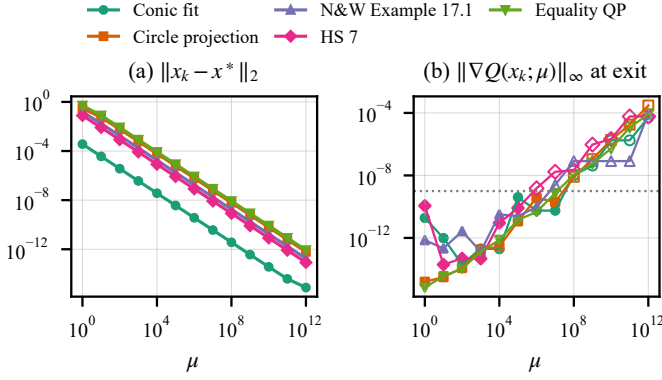


Fig. 5. Penalty method run to $\mu = 10^{12}$. (a) Distance to the solution. (b) Gradient norm of Q when the inner Newton method returns; the dotted line is the Newton tolerance 10^{-9} , and hollow markers denote subproblems that ended with the status stalled.

The relative residual of the linear solves stays below 10^{-13} , as expected from a backward stable LU factorisation, and no Hessian modification is needed except in the non-convex first subproblem of Example 17.1 ($\tau = 6.1$).

The Newton method nevertheless fails to meet its own tolerance. The gradient norm at return grows roughly in proportion to μ and reaches 5.1×10^{-5} for the conic fit at $\mu = 10^{12}$. The first stalled subproblem occurs at $\mu = 10^6$ for HS 7, at 10^7 for Example 17.1 and at 10^8 for the other problems. This matches a simple estimate: a computed minimiser can only be located to a relative accuracy of order $\varepsilon_{\text{mach}}$, so the smallest gradient norm that can be observed is about $\varepsilon_{\text{mach}} \|\nabla^2 Q\| \|x\|$. For the conic fit $\|\nabla^2 Q\|_2 \approx 4\mu$ and $\|\theta\| = 1$, so this level passes 10^{-9} at $\mu \approx 10^6$ and is about 10^{-3} at $\mu = 10^{12}$, an upper estimate for the observed 5.1×10^{-5} . The loss caused by ill-conditioning therefore appears first in the stopping tests: the inner gradient tolerance and the outer KKT tolerance become unattainable, while x_k itself remains close to the minimiser of Q .

Fig. 6 shows the same effect geometrically. For $\mu = 1$ the penalty function is not convex and its minimiser lies well outside the circle. As μ grows the level sets turn into a narrow curved valley along the constraint; its curvature across the valley grows like μ while the curvature along it stays bounded, which is the eigenvalue split behind (8). The minimiser of $\mathcal{L}_A(x; \lambda^*, 10)$ is x^* itself, and its valley keeps the width it has at $\mu = 10$.

D. Parameter sensitivity

Table III varies the growth factor of the penalty method. A larger ρ reduces the number of subproblems, from 127 for $\rho = 2$ to 19 for $\rho = 1000$, and the Newton iterations from 270 to 78, because warm starts stay effective even for large jumps in μ . The price is a less controlled final μ . With $\rho = 100$ the circle projection misses the feasibility tolerance at $\mu = 10^8$ ($\|h\|_\infty = 1.24 \times 10^{-8}$), and at $\mu = 10^{10}$ the KKT residual is 1.9×10^{-6} because the gradient can no longer be computed accurately enough. The run ends at $\mu_{\text{max}} = 10^{12}$ with $\kappa_\infty =$

TABLE III
PENALTY GROWTH FACTOR ρ WITH $\mu_0 = 1$: TOTALS OVER THE FIVE PROBLEMS, LARGEST NEWTON COUNT OF ONE PROBLEM, WORST ERROR, LARGEST CONDITION NUMBER AND CONVERGED RUNS

ρ	Subpr.	Newton	Max.	$\max \ x - x^*\ $	$\max \kappa_\infty$	Conv.
2	127	270	75	6.3×10^{-9}	7.7×10^8	5/5
5	59	150	42	4.7×10^{-9}	2.0×10^9	5/5
10	44	127	33	3.7×10^{-9}	5.7×10^9	5/5
100	27	88	25	1.8×10^{-9}	1.3×10^{12}	4/5
1000	19	78	23	8.5×10^{-10}	8.0×10^9	5/5

TABLE IV
CONIC FIT OVER 15 DATA SETS PER NOISE LEVEL (10, 40 AND 160 POINTS, FIVE SEEDS): MEAN SUBPROBLEMS, MEAN NEWTON ITERATIONS AND LARGEST κ_∞

Noise	Quadratic penalty			Augmented Lagrangian		
	Subpr.	Newton	κ_∞	Subpr.	Newton	κ_∞
0	1.0	5.0	2.5×10^1	1.0	5.1	1.0×10^2
0.02	5.8	12.5	1.0×10^6	2.0	7.7	1.0×10^2
0.05	6.3	14.3	8.5×10^6	2.0	8.2	1.0×10^2
0.1	6.9	16.6	9.9×10^6	2.0	8.3	9.8×10^1
0.2	7.8	19.2	1.0×10^8	2.0	8.3	1.1×10^2

1.3×10^{12} and without a converged status, although x_k is accurate to 6×10^{-13} .

Fig. 7 varies the fixed parameter of the augmented Lagrangian. The condition number grows in proportion to μ exactly as for the penalty method. For the four nonlinear problems the work is smallest for μ between 1 and 100. Larger values reduce the number of outer iterations, as (14) predicts, but every subproblem becomes harder: the circle projection needs 26 Newton iterations at $\mu = 10$ and 197 at $\mu = 10^6$. For Example 17.1 with $\mu = 10^6$ every subproblem ends at the Newton iteration limit, and after 30 outer iterations the iterate is still 2.76 away from x^* . The equality QP behaves differently: for a quadratic objective and linear constraints $\mathcal{L}_A$ is quadratic, Newton needs one step per subproblem, and large μ only speeds up the multiplier convergence (27 iterations at $\mu = 1, 2$ at $\mu \geq 10^4$). The adaptive rule (15) started from $\mu_0 = 1$ converges on all problems with $\mu \leq 5$ and $\kappa_\infty \leq 32$.

E. Robustness of the curve fit

Table IV repeats the curve fit for five noise levels, three data sizes and five seeds. All 150 runs converge. Without noise the data lie exactly on a conic, so $\sigma_{\min}(M) = 0$, $\lambda^* = 0$ and the penalty function is exact: one subproblem at $\mu = 1$ suffices. With noise, $|\lambda^*| = \sigma_{\min}(M)$ equals the fitting residual and grows with the noise level. By (6) the penalty method then needs a larger final μ , and its largest condition number rises from 10^6 at noise 0.02 to 10^8 at noise 0.2. For every nonzero noise level the augmented Lagrangian needs two subproblems and about eight Newton iterations, with $\kappa_\infty \approx 10^2$.

VII. DISCUSSION

The measurements confirm the classical analysis quantitatively. The growth $\kappa \propto \mu$ in (8) holds to four digits in the slope,

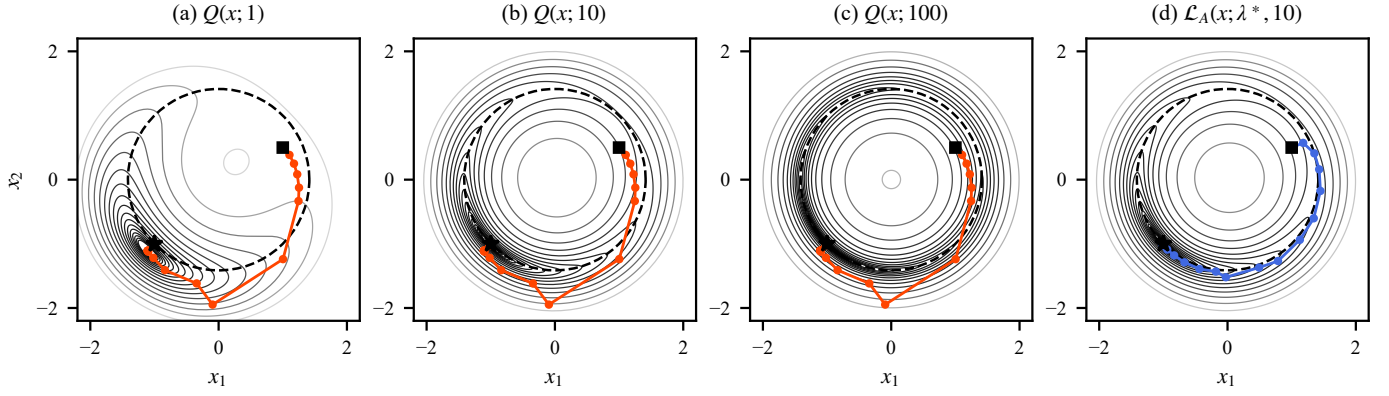


Fig. 6. Nocedal and Wright Example 17.1. (a) to (c) Contours of $Q(x; \mu)$ for $\mu = 1, 10, 100$ with the Newton iterates of all subproblems up to that μ (square: start, star: x^* , dashed: constraint). (d) Contours of $\mathcal{L}_A(x; \lambda^*, 10)$ with the augmented Lagrangian iterates.

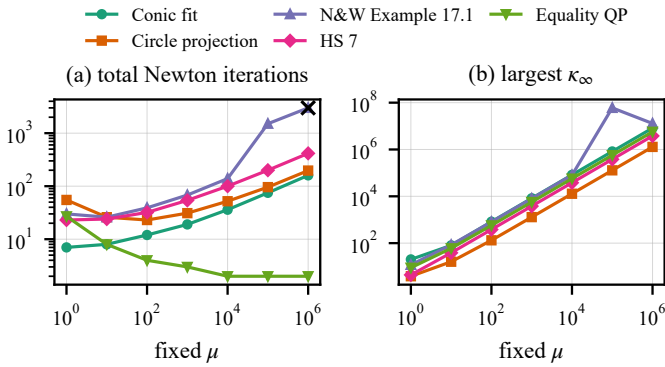


Fig. 7. Augmented Lagrangian with a fixed μ . (a) Total Newton iterations; the cross marks a run that did not converge within 30 outer iterations. (b) Largest condition number.

in all three norms and on all five problems, and the penalty violation matches λ^*/μ in (6). The augmented Lagrangian removes the need for a large μ , and with it the ill-conditioning, but it does not change the conditioning for a given μ . Choosing μ too large in the augmented Lagrangian brings back the same problems, including a failure on Example 17.1.

The effect of ill-conditioning on Newton's method is more specific than a general loss of accuracy. With exact Hessians and a backward stable linear solver, the Newton steps stay accurate and the iterates keep approaching the minimisers of Q . What fails first is the ability to verify convergence, because the smallest gradient norm that can be observed near a minimiser grows in proportion to $\|\nabla^2 Q\|$ and therefore to μ . In practice this means inner iterations that end without meeting their tolerance, extra iterations spent in the rounding regime, and outer KKT tests that cannot be met once μ exceeds about 10^{10} . With inexact Hessians, iterative linear solvers or larger problems, the same conditioning would also damage the steps themselves [4].

The penalty method is competitive when $|\lambda^*|$ is small, as the noise-free curve fit shows, or when a moderate feasibility tolerance is enough. Otherwise the augmented Lagrangian is

preferable: at default settings it needs fewer Newton iterations on every problem and reaches a comparable or smaller constraint violation.

The study has limitations. The problems are small and dense, so condition numbers are computed exactly through an explicit inverse or an eigenvalue decomposition, which costs $O(n^3)$ per evaluation. Only equality constraints are treated; inequality constraints would require slack variables or an active-set strategy. Timings are of the order of milliseconds and are not used for comparison.

VIII. CONCLUSION

A from-scratch implementation of the quadratic penalty and augmented Lagrangian methods with a common modified Newton solver was used to measure the conditioning of every Newton step. On five test problems the condition number of the penalty Hessian grows linearly in μ up to 8.2×10^{12} , while the augmented Lagrangian stays below 84 with $\mu = 10$ and needs fewer Newton iterations. The main practical consequence of the ill-conditioning is that inner and outer convergence tests become unattainable beyond $\mu \approx 10^6$ to 10^8 , even though the iterates remain accurate. The application built on natID makes these effects interactive: every parameter can be changed with a slider, and the landscape, convergence and conditioning views update immediately.

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